

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 34)

Suggested Solution

Paper 1

1. D	11. B	21. D	31. D	41. B
2. C	12. D	22. C	32. B	42. C
3. D	13. B	23. B	33. D	43. D
4. A	14. D	24. C	34. D	44. A
5. A	15. B	25. C	35. A	45. C
6. B	16. C	26. A	36. D	
7. C	17. A	27. B	37. A	
8. C	18. C	28. C	38. A	
9. D	19. A	29. D	39. A	
10. B	20. A	30. B	40. B	

Explanation

1. Answer: D

Tracy would choose the first option as the benefit of choosing the first option (the value of the first option) is higher than the cost (the value of the second option).

Option D is correct. This would reduce the value of the third option, without affecting Tracy's cost or benefit of choosing the first option.

Option A is incorrect. This would reduce the value of the second option, thus reducing Tracy's cost of choosing the first option.

Option B is incorrect. This may increase the value of the third option to a level higher than that of the second option, so the cost of choosing the first option may change from the value of the second option to that of the third option.

Option C is incorrect. This would increase the value of the first option, thus increasing Tracy's benefit of choosing the first option.

2. Answer: C

Mr. Chan switching to work for company B will have to give up working for company A. Therefore, the monthly salary of \$25,000 from company A is an opportunity cost of switching to work for company B.

Option D is incorrect. The tuition fee of \$100,000 for studying the human resources course is the cost that has been paid in the past and cannot be recovered, which is not the opportunity cost of switching to work for company B.

3. Answer: D

Deciding to implement e-Government services is related to what kinds of resources are used to produce services, the problem of 'how to produce' is thus related.

The decision is made by the government, government command is thus used.

4. Answer: A

(2) is correct. Sole proprietorship lacks lasting continuity. If the owner goes bankrupt, the partnership will be dissolved. A limited company is a legal entity, when one of the owners goes bankrupt, the firm will continue to exist.

(3) is incorrect. Only a public limited company can issue shares to the public, a private limited company cannot.

5. Answer: A

Option B is incorrect. Private goods are not necessarily provided by private enterprises, e.g. the recreation and sports facilities provided by the Leisure and Cultural Services Department are private goods.

Option C is incorrect. Free television programmes produced by private television stations is non-rival in consumption. The consumption of the programmes by one person does not reduce the amount available to others.

Option D is incorrect. Directly satisfying human wants is a characteristic of consumer goods, not private goods.

6. Answer: B

Occupational mobility refers to the ability and willingness of a factor of production to change from one occupation to another. Compared with the other three people, a principal has the highest salary and is required to have the highest level of professional knowledge, so the opportunity cost of changing occupations will be higher. As a result, a principal is the least mobile in terms of occupational mobility.

7. Answer: C

Professionals are generally more productive. More foreign professionals migrating to work in Country A will raise the labour supply and average labour productivity of Country A.

(2) is incorrect. This will only raise the geographical mobility of foreign professionals and will not affect the geographical mobility of workers in Country A.

8. Answer: C

Labour (units)	Machines (units)	Average product (units)	Total product (units)	Marginal product (units)
2	10	6	12	-
3	10	9	27	15
4	10	12	48	21
5	10	14	70	22
6	10	15	90	20

When the 6th unit of labour is employed, marginal product decreases from 22 units to 20 units.

9. Answer: D

Output (units)	Total cost (\$)	Marginal cost (\$)	Total cost after an increase in fixed cost (\$)	Marginal cost after an increase in fixed cost (\$)
1	5	4	7	4
2	11	6	13	6
3	19	8	21	8
4	29	10	31	10
5	41	12	43	12

The original market price is \$12 and the profit-maximising output is 5 units.

Producer surplus = $(\$12 \times 5) - (\$4 + \$6 + \$8 + \$10 + \$12) = \$20$

The new market price is \$8 and the profit-maximising output is 3 units.

Producer surplus = $(\$8 \times 3) - (\$4 + \$6 + \$8) = \$6$

The profit-maximising output decreases by 2 units and the producer surplus decreases by \$14.

10. Answer: B

(1) is inconsistent. Reducing the unemployment benefits will increase the opportunity cost of people being unemployed, the number of unemployed people will decrease.

(3) is inconsistent. The relative price of a long stay (in terms of a short stay) will decrease. According to the law of demand, a higher proportion of visitors will choose to stay longer. The average time of people staying in the swimming pools will increase.

(2) is consistent. The law of demand states that an increase in the price of a good will result in a decrease in its quantity demanded, *ceteris paribus*. Cold spells are one of the factors other than the price that affects the demand for heaters. As the demand for heaters increases, both their quantity transacted and price will increase.

11. Answer: B

Wooden furniture and wood chips are in joint supply. When the price of wooden furniture decreases, its quantity supplied will decrease, and the supply of wood chips will also decrease.

Option A is incorrect. Flour is the factor of production of bread. When the price of flour decreases, the production cost of bread will decrease, and the supply of bread will increase.

Option C is incorrect. Tutors are the factors of production needed to provide tutoring services. The change in the price of tutorial services will not affect the supply of tutors.

Option D is incorrect. Cross-harbour bus service and cross-harbour rail service are in competitive demand. When the price of cross-harbour bus service decreases, the demand for cross-harbour rail service will decrease, but its supply will not be affected.

12. Answer: D

The diagram shows that both the demand for and supply of private courier services increase.

The increase in the use of online shopping services will increase people's demand for courier services, and the introduction of artificial intelligence to assist in the distribution of mail will increase the supply of courier services, causing the equilibrium point to move from E_0 to E_1 .

(1) is incorrect. Courier services provided by the government postal institution and courier services provided by private courier companies are substitutes. When the government postal institution lowers the courier service charges, the demand for private courier services will decrease.

(2) is incorrect. Many couriers switching to other industries will lead to a decrease in the supply of courier services.

13. Answer: B

More people believing that taking vitamin pills can help maintain good health will cause the demand for vitamin pills to increase. Both the price and quantity transacted will increase. According to the information in the question, since the percentage change in price (10%) is smaller than the percentage change in quantity supplied (50%), the supply of vitamin pills is elastic.

When vitamin pill manufacturers have spare production capacity, the supply of vitamin pills will tend to be elastic.

Option A is incorrect. When the government sets up strict licensing requirements for vitamin pill manufacturers, new manufacturers cannot enter the market easily, the supply of vitamin pills will tend to be inelastic.

Options C and D are factors that affect the price elasticity of demand and cannot explain the changes in price and quantity supplied.

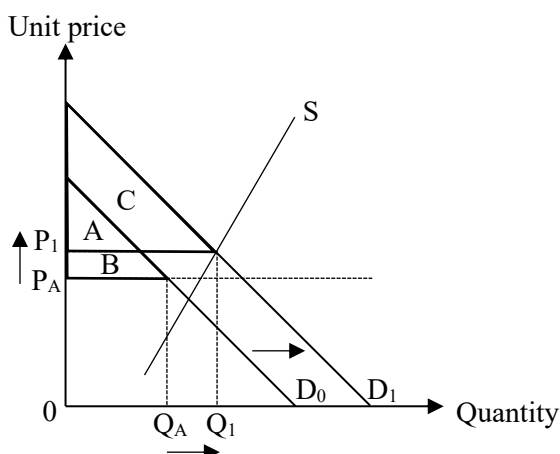
14. Answer: D

(3) is correct. If the demand for batteries is more elastic than the supply, producers' share of the subsidy will be larger than consumers' share of the subsidy. Consumers' share of the subsidy per unit of batteries will be smaller than \$3. Therefore, the equilibrium price after the provision of the per-unit subsidy can be \$29 (Consumers' share of the subsidy per unit of batteries = \$1).

(4) is correct. If the demand for batteries is perfectly elastic or the supply of batteries is perfectly inelastic (demand is more elastic than supply), after the provision of the per-unit subsidy, the equilibrium price will remain unchanged, i.e. \$30.

(1) and (2) are incorrect. As mentioned above, consumers' share of the subsidy per unit of batteries will be smaller than \$3, therefore, the new equilibrium price cannot be \$24 or \$27.

15. Answer: B



Before the increase in demand, the price floor P_A is effective, the quantity transacted is Q_A . There is a surplus in the market of good X.

After the demand increases from D_0 to D_1 , the price floor P_A becomes ineffective. The new price is P_1 and the quantity transacted is Q_1 .

(1) is correct. The total expenditure increases from $(P_A \times Q_A)$ to $(P_1 \times Q_1)$.

(3) is correct. The consumer surplus will increase from area $(A + B)$ to area $(A + C)$.

(2) is incorrect. After the price floor becomes ineffective, the market of good X is in equilibrium.

16. Answer: C

Unit price (\$)	10	11	12	13	14	15
Quantity demanded (units)	45	40	35	30	25	20
Quantity supplied (units)	25	30	35	40	45	50
Quantity supplied after tax (units)	-	-	25	30	35	40

Before tax, the equilibrium price of good X is \$12 and the quantity transacted is 35 units.

After tax, the equilibrium price of good X is \$13 and the quantity transacted is 30 units.

(2) is correct. Before tax, producers' total revenue is \$420. After tax, producers' total revenue net of tax is \$330 $(= (\$13 - \$2) \times 30)$.

(3) is correct. Consumers' tax burden $= (\$13 - \$12) \times 30 \text{ units} = \30

Producers' tax burden $= [\$12 - (\$13 - \$2)] \times 30 \text{ units} = \30

Consumers and producers bear the same tax burden.

(1) is incorrect. Before tax, consumers' total expenditure on good X is \$420. After tax, the price of good X increases and consumers' total expenditure on good X decreases to \$390. This indicates that the percentage increase in price is smaller than the percentage decrease in quantity demanded, and the demand for good X is elastic.

17. Answer: A

Option B is incorrect. There may be many buyers in both perfectly competitive market and imperfectly competitive market, so this cannot be used to explain why the market structure of cha chaan tengs in Hong Kong is not perfectly competitive.

Options C and D are incorrect. They are also features of a perfectly competitive market.

18. Answer: C

(2) is correct. A decrease in the marginal benefit per unit of rice means that there is a decrease in demand for rice. The price and quantity transacted will decrease, and producer surplus will decrease.

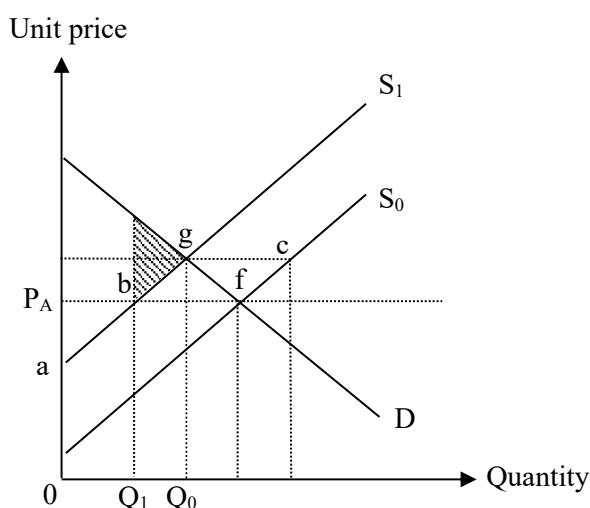
(3) is correct. Producers will supply rice until the price equals marginal cost. When the demand for rice decreases, its price will decrease and the marginal cost of the last unit of rice will decrease.

(1) is incorrect. When demand decreases, total social surplus will decrease, but this does not mean that economic efficiency will be worsened. Before and after the demand changes, at the equilibrium quantity, the marginal benefit is equal to the marginal cost, and the total social surplus reaches the maximum, i.e. economic efficiency is achieved.

19. Answer: A

When the supply of good X decreases from S_0 to S_1 and the government sets a price ceiling at P_A , the price ceiling will be lower than the new equilibrium price and the price ceiling is effective. The output will equal to Q_1 and producer surplus is P_Aba .

At Q_1 , output is lower than the efficient output level (Q_0), i.e. there is under-production. Marginal benefit is greater than marginal cost, there is a deadweight loss, which is the shaded part in the diagram below.



20. Answer: A

Stores playing advertisements loudly and repeatedly without paying any compensation to nearby residents, external cost exists and there is a divergence between private and social costs.

(1) is correct. Assume there is no divergence between private and social benefits. Since the marginal private cost is lower than the marginal social cost, the quantity of advertisements played (i.e. the level where marginal private cost = marginal private benefit) will be higher than the efficient level (i.e. the level where marginal social cost = marginal social benefit).

(2) is correct. The marginal private cost of playing advertisements for these stores will increase and they will reduce the quantity of advertisements played. The divergence between private and social costs will be reduced.

(3) is incorrect. Government imposing fines on these stores will increase the marginal private cost of playing advertisements for these stores. The divergence between private and social costs will be reduced, but not necessarily eliminated.

21. Answer: D

(1) is correct. The disposable income of the recipients will increase. As the recipients are mainly low-income people, the income gap between the rich and the poor will narrow. The Gini coefficient based on household disposable income will decrease.

(2) is correct. Increasing the basic allowance for the salaries tax will reduce the number of low-income people who need to pay salaries tax. The income gap between the rich and the poor will narrow. The Gini coefficient based on household disposable income will decrease.

(3) is incorrect. Municipal solid waste charging is a regressive tax because the same amount of charging increase accounts for a larger proportion of the income of the low-income group than that of the high-income group. Increasing the amount of municipal solid waste charging will widen the income gap between the rich and the poor. The Gini coefficient based on household disposable income will increase.

22. Answer: C

The difference in natural abilities among insurance agents tends to be greater than that of waiters. More talented insurance agents may earn an income much higher than the average income. The difference in natural abilities among waiters is smaller, so the income difference among waiters tends to be smaller.

23. Answer: B

When calculating Hong Kong's GDP, those living in owner-occupied units are assumed to provide rental services to themselves, so estimated rental value of an owner-occupied unit would be included in the calculation of Hong Kong's GDP.

Options A and C are incorrect. Neither the funds raised from flag selling nor capital gains are generated from production and therefore would not be included in the calculation of GDP.

Option D is incorrect. The residential unit was not produced in the current year and its value was included in the calculation of GDP of the year when it was produced.

24. Answer: C

The contribution of the production chain to Country A's GDP at market prices
= (\$9,000 – \$6,000) + (\$5,000 + \$6,000 – \$500 – \$600) + \$600
= \$13,500

25. Answer: C

Growth rate of real GDP is 1%, reflecting an increase in real GDP.

Growth rate of population is negative, reflecting a decrease in population.

Per capita nominal GDP is 0% (i.e. no change), meaning that nominal GDP and population has changed at the same percentage (–3%). Nominal GDP has decreased while real GDP has increased (1%), implying a decrease in the general price level.

As the population has decreased and the real GDP has increased, per capita real GDP has increased.

26. Answer: A

(1) is correct.

Gross domestic fixed capital formation = Net domestic fixed capital formation + Depreciation

As depreciation (i.e. capital consumption) is 10 million, the gross domestic fixed capital formation is larger than the net domestic fixed capital formation.

(2) is correct.

GNI = GDP + Net factor income from abroad

As net factor income from abroad is positive, GNI is greater than GDP.

(3) is incorrect. GDP at market prices = GDP at factor cost + Indirect taxes – Subsidies

As indirect taxes are smaller than subsidies, GDP at market prices is smaller than GDP at factor cost.

27. Answer: B

When people's desire to save increases, private consumption expenditure will decrease. Aggregate demand will decrease and the price level in the short run will decrease.

Option B is correct. When the government raises the minimum wage rate, the production cost will increase and the short run aggregate supply will decrease. The price level in the short run will increase.

Option A is incorrect. When producers expect the price level to decrease, the short run aggregate supply will increase and the price level in the short run will decrease further.

Option C is incorrect. When the risk of holding interest-bearing assets increases, money demand will increase and the interest rate will increase. Consumption and investment will decrease. Aggregate demand will decrease and the price level in the short run will decrease further.

Option D is incorrect. When the economy's trading partners experience economic recession, the economy's net exports will decrease. Aggregate demand will decrease and the price level in the short run will decrease further.

28. Answer: C

An increase in government expenditure will increase aggregate demand.

As the productive capacity increases, both the short run and the long run aggregate supply will increase. Potential output will increase in the long run.

The price level in the long run may not increase, depending on the relative extent of the increase in aggregate demand and the increase in the long run aggregate supply.

29. Answer: D

Option A is incorrect. The Budget of the Hong Kong government is prepared by the Financial Secretary.

Option B is incorrect. The \$100 banknotes are issued by the three note-issuing banks in Hong Kong.

Option C is incorrect. The Hong Kong Monetary Authority does not provide loans to the public.

30. Answer: B

Withdrawing HK\$500,000 from an account in a restricted license bank will cause M1 and M2 to increase by HK\$500,000 without affecting M3.

Holding HK\$100,000 as cash will not affect M1, M2 and M3.

Depositing HK\$200,000 into a time deposit account in a licensed bank will cause M1 to decrease by HK\$200,000 without affecting M2 and M3.

Remitting the remainder of HK\$200,000 overseas will cause M1, M2 and M3 to decrease by HK\$200,000.

In conclusion, M1 will increase by HK\$100,000, M2 will increase by HK\$300,000, and M3 will decrease by HK\$200,000.

31. Answer: D

Required reserve ratio = $(\$400 \text{ million} / \$1,600 \text{ million}) \times 100\% = 25\%$

After the public withdraw \$100 million from the banking system,

Maximum possible amount of deposits = $(\$400 \text{ million} - \$100 \text{ million}) \times (1/25\%) = \$1,200 \text{ million}$

Maximum possible amount of loans = $\$1,200 \text{ million} - \$300 \text{ million} = \$900 \text{ million}$

Option A is incorrect. Monetary base = $(\$400 \text{ million} - \$100 \text{ million}) + \$100 \text{ million} = \400 million

Option B is incorrect. Money supply = $\$1,200 \text{ million} + \$100 \text{ million} = \$1,300 \text{ million}$

32. Answer: B

Monetary base = Cash in public circulation + Bank reserves

Money supply = Cash in public circulation + Bank deposits

(2) is correct. When the central bank lowers the required reserve ratio, the banks will have more excess reserves and they can make more loans to the public. Through deposit creation, deposits will increase and money supply will increase. On the other hand, lowering the required reserve ratio will not cause the monetary base to change, as cash in public circulation and bank reserves do not change.

(1) is incorrect. When the central bank lowers the discount rate, commercial banks can then borrow money from the central bank at a lower cost. With more reserves, they can make more loans to the public. Monetary base will increase. Through deposit creation, deposits will increase and money supply will increase.

(3) is incorrect. When the central bank purchases government bonds from the public, the money that the central bank pays out will circulate in the public or will be deposited in banks. The increase in the cash in public circulation directly increases the money supply while the increase in the money deposited in banks increases the cash reserves of the banking system. Through deposit creation, money supply will increase further. The increase in cash in public circulation or bank reserves will increase the monetary base.

33. Answer: D

When people are more used to making payments for daily expenses by e-wallets, they will hold less money for transactions. Transaction demand for money will decrease.

When the central bank sells government bonds to the public, the money supply will decrease.

When the money demand and the money supply decrease at the same time, the quantity of money will decrease and the nominal interest rate may increase or decrease, depending on the relative effects of the decrease in money demand and the decrease in money supply on the nominal interest rate.

34. Answer: D

When there is unexpected deflation, the real value (purchasing power) of the interest paid by the bond issuers will increase. The bond issuers will thus lose.

Option A is incorrect.

Actual real interest rate = Nominal interest rate – Actual inflation rate = $4\% - (-3\%) = 7\%$

Option B is incorrect. Cost of holding money = Nominal interest rate = 4%

Option C is incorrect. The question does not provide information on the expected real interest rate.

35. Answer: A

Quantity theory of money: $MV = PY$, where M = money supply, V = velocity of circulation of money, P = price level and Y = real output, assuming V is constant.

$\% \text{ change in } M \approx \% \text{ change in nominal GDP} \approx \% \text{ change in } P + \% \text{ change in } Y$

When the growth rate of M is positive and constant, and the growth rate of P changes from a growth rate lower than that of M to a growth rate higher than that of M , the percentage change in Y changes from positive to negative. The growth rate of real output is decreasing.

Option B is incorrect. The inflation rate is equal to the growth rate of the general price level. The diagram shows that the inflation rate is increasing.

Option C is incorrect. The quantity theory of money assumes that the velocity of circulation of money is a constant.

Option D is incorrect. The growth rate of the money supply is positive, meaning that the money supply is increasing.

36. Answer: D

When many civil servants retired, the labour force would decrease. If the government has decided to fill only part of the job vacancies generated by the retirement wave, the number of unemployed population would fall, but the decrease would be smaller than the decrease in the labour force. The change in the unemployment rate would be uncertain, depending on the percentage decrease in the unemployed population and the percentage decrease in the labour force. If the percentage decrease in the unemployed population is greater than the percentage decrease in the labour force, the unemployment rate would decrease. If the percentage decrease in the unemployed population is lower than the percentage decrease in the labour force, the unemployment rate would increase. If the percentage decrease in the unemployed population is equal to the percentage decrease in the labour force, the unemployment rate would remain unchanged.

37. Answer: A

Both low-income people and high-income people pay the same amount of Air Passenger Departure Tax. Raising the tax will cause the tax payment account for a higher percentage increase in the income for low-income people than for high-income people.

38. Answer: A

The old tax system was a proportional tax. The new tax system is a regressive tax and the tax allowance is cancelled.

Under the new tax system, as there is no tax allowance, more low-income people will fall into the tax net, while high-income group will pay a lower tax payment, income distribution will become more uneven.

Option B is incorrect. Since there is no tax allowance under the new tax system, more people will fall into the tax net.

Option C is incorrect. For high-income people, as the tax payment decreases for the same income, their working incentive may increase.

Option D is incorrect. There is insufficient information to determine whether a fiscal surplus will be resulted.

39. Answer: A

When the government implements a contractionary monetary policy, aggregate demand will decrease and aggregate output will decrease.

The abolition of the general sales tax will reduce the production cost, the short run aggregate supply will increase. Aggregate output will increase, which can offset the effect of a contractionary monetary policy on aggregate output.

Option B is incorrect. Raising the profits tax rate will reduce firms' investment, investment expenditure will decrease. Aggregate demand will decrease. Aggregate output will decrease further.

Option C is incorrect. When the government budget surplus increases, government tax revenue is greater than government expenditure. The effect on aggregate demand is contractionary. Aggregate output will decrease further.

Option D is incorrect. When the central bank sells government bonds to the public, the money supply will decrease. The interest rate will increase. Consumption/investment will decrease. Aggregate demand will decrease. Aggregate output will decrease further.

40. Answer: B

Reducing the profits tax rate will increase investment expenditure. Aggregate demand will increase and aggregate output will increase. Since most people who pay profits tax are high-income people, reducing the profits tax rate will widen the income gap and result in a deterioration in equity.

Option A is incorrect. Distributing HK\$20,000 to parents for each newborn baby will increase people's disposable income. Private consumption expenditure will increase and aggregate demand will increase. Aggregate output will increase. Besides, since everyone can benefit regardless of their income, the HK\$20,000 accounts for a higher percentage of income of the low-income group. Income distribution will become more even and result in an improvement in equity.

Option C is incorrect. The construction of a new railway spur line will increase government expenditure and aggregate demand will increase. At the same time, the construction of a new railway spur line will also increase the long run aggregate supply and aggregate output will increase. On the other hand, a better transportation network may reduce the cost for low-income people to work farther away from their residential areas, increasing their chances of finding jobs with higher income. An improvement in equity will be resulted.

Option D is incorrect. A provision of electronic learning devices implies an increase in government expenditure. Aggregate demand will increase. If students have more learning devices, productivity will increase in the long term. The long run aggregate supply and aggregate output will increase. Besides, providing electronic learning devices to students from low-income families allows them to receive more education, thereby improving their future productivity and income-earning ability, preventing them from being deprived of education because of their backgrounds, resulting in an improvement in equity.

41. Answer: B

When the liquidity of interest-bearing assets decreases, people will hold less of these interest-bearing assets and hold more money. Asset demand for money will increase and the interest rate will increase. Consumption and investment expenditure will decrease. Aggregate demand will decrease and the output level will decrease.

Option B is correct. Lowering the required reserve ratio by the central bank will increase the money supply. The interest rate will decrease, which will stimulate consumption and investment expenditure. Aggregate demand will increase and the output level will increase, thus stabilising the output level and the interest rate in the economy.

Option A is incorrect. When the central bank raises the discount rate, the money supply will decrease and the interest rate will increase further.

Option C is incorrect. When the government raises the salaries tax rate, people's disposable income will decrease and private consumption expenditure will decrease. Aggregate demand will decrease and the output level will decrease further.

Option D is incorrect. When the government increases the time interval between income payments for civil servants, people will increase the amount of money they hold for transactions. Transaction demand for money will increase and the interest rate will increase further.

42. Answer: C

Country A can use all its resources to produce 40 units of computers or 120 units of bread while

Country B can use all its resources to produce 80 units of computers or 110 units of bread.

Country A's opportunity cost of producing 1 unit of computers = 3 ($= 120/40$) units of bread

Country B's opportunity cost of producing 1 unit of computers = 1.375 ($= 110/80$) units of bread

Since Country B has a lower opportunity cost in producing computers, Country B will specialise in producing computers, while Country A will specialise in producing bread.

Before specialisation, the total output of computers is 60 units and the total output of bread is 115 units.

After specialisation, the total output of computers is 80 units and the total output of bread is 120 units.

The total output of computers will increase by 20 units, while the total output of bread will increase by 5 units.

43. Answer: D

(1) is correct. Both the domestic price and consumption of good X will increase, so the total expenditure of local consumers on good X will increase.

(2) is correct. The change in import volume depends on the relative extent of the increase in the world price and the increase in demand. As the change in import volume is uncertain, so the change in tariff revenue is uncertain.

(3) is correct. As the domestic price increases, the domestic production of good X will increase.

44. Answer: A

(1) is correct. This will be included in the imports of services.

(2) is correct. This will be included in the current transfer received from abroad.

(3) is incorrect. This will be included in the calculation of the capital and financial account.

45. Answer: C

More Japanese yen can be exchanged for each unit of the US dollar, the US dollar appreciates against the Japanese yen. Under the linked exchange rate system, the Hong Kong dollar also appreciates against the Japanese yen.

Option C is correct. The depreciation of the Japanese yen against the Hong Kong dollar makes the goods and services in Hong Kong more expensive (in terms of Japanese yen) for Japanese tourists, resulting in a decrease in the number of Japanese tourists visiting Hong Kong.

Option A is incorrect. The Japanese yen depreciates against the US dollar. Suppose the price of imports from Japan to the United States (in terms of Japanese yen) remains unchanged. The price of imports in terms of the US dollar will decrease. Import volume will increase and the change in total import value (in terms of the US dollar) will be uncertain, depending on United States's price elasticity of demand for Japanese goods.

Option B is incorrect. When the Japanese yen depreciates against the Hong Kong dollar, assuming that the price of goods exported from Hong Kong to Japan (in terms of the Hong Kong dollar) remains unchanged, the price of exports in terms of Japanese yen will increase. Japan's demand for Hong Kong's goods will decrease and Hong Kong's exports of goods to Japan will decrease. The total value of goods exported from Hong Kong to Japan (in terms of the Hong Kong dollar) will decrease.

Paper 2

Section A (44 marks)

1. (a) To Mr. Chan, the light and the installation service provided by him were the price/cost for obtaining the \$30,000 earlier. (2 marks)
- (b) The opportunity cost will not change as the fact that Miss Wong does not like the light will only lower the value of Mr. Chan installing the new light for Miss Wong's home, but does not affect Mr. Chan's highest-valued option forgone. (2 marks)
2. (a) Firm A's average labour productivity = $900 \text{ units} / (15 \times 6 \text{ man-hours}) = 10 \text{ units/man-hour}$ (1 mark)
- Firm B's average labour productivity = $975 \text{ units} / (10 \times 6.5 \text{ man-hours}) = 15 \text{ units/man-hour}$ (1 mark)
- Therefore, firm B has a higher average labour productivity. (1 mark)
- (b) Compared to firm B, firm A has a higher percentage of output with quality up to standard, implying that workers may not need to rush for more output at the expense of quality. Thus, firm A is more likely to pay its workers at time rate. (2 marks)
- (c) Possible reasons:
- Firm B provides more training for its workers.
 - The working condition in firm B is better.
 - Better benefits and rewards are provided in firm B.
 - Firm B uses more advanced capital goods.
 - The organisation and management of labour is better in firm B. For example, division of labour is practised.
 - Workers of firm B are healthier.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- 1@, max: 2 marks
3. (a) Reasons:
- Mr. Chan would like to invite more investors to obtain new capital for business expansion.
 - Mr. Chan can make use of different expertise from Miss Li to provide tutorial service for more subjects.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- 1@, max: 2 marks
- (b) Possible reasons:
- Different branches can share teaching materials, so the cost of producing teaching materials can be spread over a larger quantity of output.
 - A large-scale tutorial center can practise a higher degree of division of labour, such as hiring specialists to work different duties.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- 2@, max: 4 marks

4. (a) Yes, because it can be refutable by facts. (2 marks)

(b) Verbal elaboration:

The effective minimum wage level will be set above the equilibrium. Raising the minimum wage level will lead to a decrease in the quantity transacted of cleaners. (2 marks)

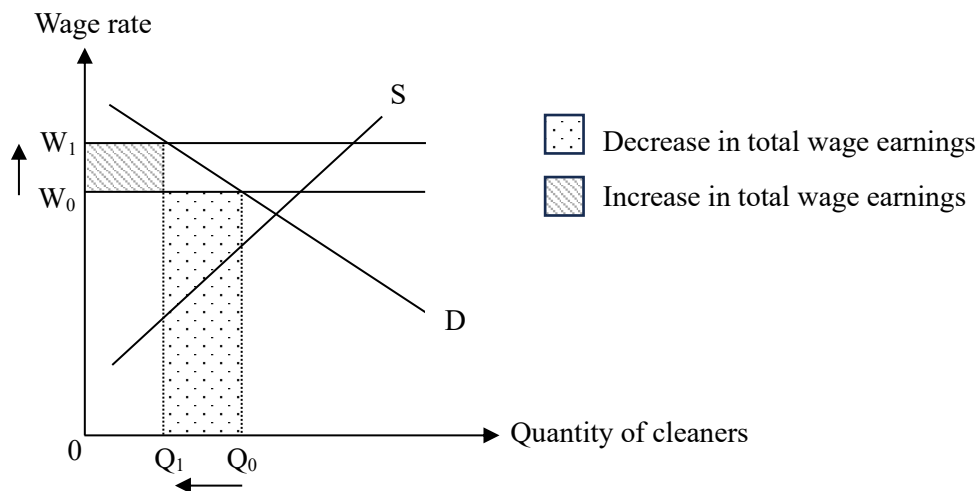
If the demand for cleaners is elastic, the percentage decrease in quantity demanded will be greater than the percentage increase in the wage rate. Raising the minimum wage level will result in a reduction in the total wage earnings of cleaners. (2 marks)

Illustrate in the diagram:

- an increase in the minimum wage level (1 mark)

- correct positions of decrease in total wage earnings and increase in total wage earnings (1 mark)

- decrease in total wage earnings > increase in total wage earnings (1 mark)



5. (a) Inflation rate = $(102 - 108) / 108 \times 100\% = -5.56\%$ (1 mark)

(b) A difference:

- CPI covers only a basket of consumer goods purchased by a typical household in an expenditure group, while GDP deflator includes all final goods and services (both consumer goods and producer goods).
- CPI is calculated using fixed weights based on consumption patterns of different expenditure groups, while GDP deflator is calculated using variable weights based on the domestic production patterns of goods and services.

2@, max: 2 marks

- Any reasonable answers

(Mark the **FIRST** point only)

(c) The decrease in the consumer price index indicates a decrease in the general price level.

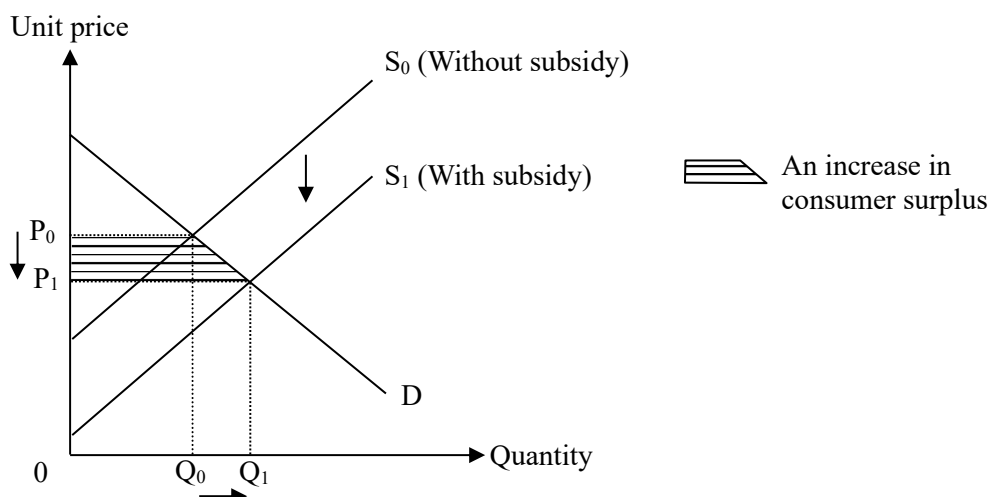
Quantity theory of money: $MV = PY$, where M = money supply, V = velocity of circulation of money, P = price level and Y = real output, assuming V is constant.

When the percentage increase in M is smaller than the percentage increase in Y , the percentage change in P will be negative, i.e. the general price level will decrease. (4 marks)

6. (a) Country A's opportunity cost of producing 1 unit of watches = $1.67 (= 20 / 12)$ units of computers (1 mark)
- Country B's opportunity cost of producing 1 unit of watches = $3 (= 15 / 5)$ units of computers (1 mark)
- Country A has a lower opportunity cost in producing watches, so Country A will specialise in producing watches and export watches. (1 mark)
- (b) Terms of trade: 1 unit of watches = 2.5 units of computers
- Country A's gain per unit of watches exported = $2.5 \text{ units of computers} - 1.67 \text{ units of computers} - 0.2 \text{ units of computers} = 0.63 \text{ units of computers}$ (3 marks)
7. (a) Balance of current account = $\$(100 - 20 + 55 - 50) \text{ million} = \85 million (1 mark)
- Since the balance of the current account is positive, the current account is in surplus. (1 mark)
- (b) Balance of payments = Balance of current account + Balance of capital and financial account (excluding changes in reserve assets)
- $= \$[85 + (-120)] \text{ million} = -\35 million (1 mark)
- There is a balance of payments deficit. (1 mark)
- Reserve assets decreases by \$35 million. (1 mark)

Section B (60 marks)

8. (a) Negative externality. The use of diesel private cars will cause air pollution, which will bring costs to others (such as damage to health), and the drivers of diesel private cars do not give any compensation. An external cost is imposed on others. The marginal social cost is higher than the marginal social benefit. The quantity of diesel private cars is higher than the efficient level. (3 marks)
- (b) Verbal elaboration:
- The supply will increase. (1 mark)
- The price paid by consumers will decrease. (1 mark)
- The quantity transacted will increase. (1 mark)
- Consumer surplus will increase. (1 mark)
- Illustrate in the diagram:
- a downward shift of the supply curve (1 mark)
 - price decreases and quantity transacted increases (1 mark)
 - correct position of an increase in consumer surplus (1 mark)



- (c) When a fixed subsidy is provided for each bicycle rental, the charge will be reduced by the same amount regardless of the rental time. The relative price of short-rental time (in term of long-rental time) will decrease. According to the law of demand, a higher proportion of people will take short-rental time, causing the average rental time for people using bicycles to decrease. (4 marks)

9. (a) A possible reason:

- Banks may deliberately keep more reserves than required to prepare for sudden withdrawals of funds by a large number of depositors.
- There is insufficient demand for loans.
- Any reasonable answer

1@, max: 1 mark

(Mark the **FIRST** point only)

(b) Original money supply = \$8,000 million + \$500 million = \$8,500 million

Required reserve ratio = $[(\$3,000 \text{ million} - \$1,000 \text{ million}) / \$8,000 \text{ million}] \times 100\% = 25\%$

New money supply = $(\$3,000 \text{ million} + \$20 \text{ million}) \times (1/25\%) + \$500 \text{ million} = \$12,580 \text{ million}$

Maximum possible change in money supply = $\$12,580 \text{ million} - \$8,500 \text{ million} = \$4,080 \text{ million}$ (4 marks)

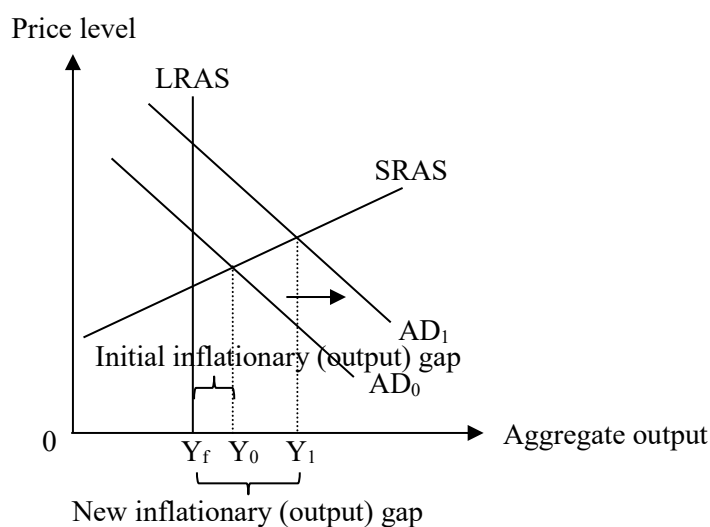
(c) (i) Inflationary (output) gap refers to the difference by which the short run equilibrium output level is in excess of the full-employment output level. (1 mark)

(ii) Verbal elaboration:

The central bank purchasing government bonds from the public would cause the money supply to increase, resulting in a decrease in the interest rate. Consumption and investment expenditure will increase and aggregate demand will increase. Aggregate output in the short run will increase, widening the inflationary (output) gap. (4 marks)

Illustrate in the diagram:

- correct initial inflationary (output) gap (1 mark)
- rightward shift of the AD curve (1 mark)
- correct position of new short-run equilibrium aggregate output Y_1 with a wider inflationary (output) gap (1 mark)



10. (a) The figure shows that USD depreciated against GBP. Under the Linked Exchange Rate System of Hong Kong, HKD would also depreciate against GBP. (2 marks)
- (b) As HKD depreciated against GBP, the tuition fees and living expenses (in terms of HKD) for Hong Kong students studying in Britain would increase. Therefore, the cost of living in terms of Hong Kong dollars for Hong Kong students studying in Britain would increase. (2 marks)

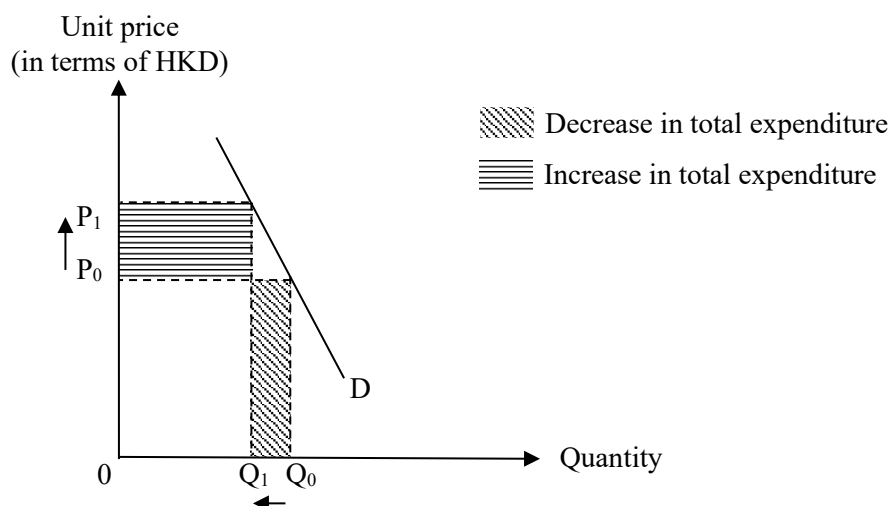
(c) Verbal elaboration:

As HKD depreciated against GBP, the price of goods imported from Britain to Hong Kong (in terms of HKD) would increase.

If Hong Kong's demand for British goods is inelastic, the percentage decrease in quantity demanded will be smaller than the percentage increase in price. The total expenditure of Hong Kong on British goods (in terms of HKD) will increase. (5 marks)

Illustrate in the diagram:

- correct positions of the increase in total expenditure and decrease in total expenditure (1 mark)
- increase in total expenditure > decrease in total expenditure (1 mark)



11. (a) Monopoly. (1 mark)
- This is because there is only one power company providing electricity to each district. (1 mark)
- (b) Yes, because the electricity charges relief reflects the value of goods produced by resident producing units in the current period. (2 marks)
- (c) According to Source C, electricity, gas and water have a relatively small weight (2.82%) in the Composite CPI. Increasing electricity charges will not significantly raise the Composite CPI, resulting in a minor increase in the cost of living. (2 marks)
- (d) Demand for solar power systems will increase. (1 mark)
- The price of solar power systems will increase. (1 mark)

(e) Marks awarded for effective communication (EC: max 2 marks)

Marks	Performance
2	• <i>Arguments supported with the source/data and appropriate economic theories</i> • <i>Relevant materials presented</i> • <i>Well-organised and coherent answers presented without repetition of ideas</i> • <i>Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols</i>
1	• <i>Arguments presented with some support of the source/data and economic theories</i> • <i>Some irrelevant materials presented</i> • <i>Answers presented in a less organised way with some repetition of ideas</i> • <i>A clear message conveyed with some inappropriate use of language/words/terms/symbols</i>
0	• <i>Arguments presented with no support of the source/data and economic theories</i> • <i>Materials unrelated to the gist of the question presented</i> • <i>Inconsistent arguments presented</i> • <i>Limited ideas expressed with inappropriate use of words/terms/symbols</i>

The maximum marks for content are 12 marks. Answers may include the following:

Effects on income distribution

Proposal I

- Refer to Source B. As all eligible residential electricity accounts can receive a monthly electricity charges relief, there will be a larger percentage increase in the income of low-income households. Income distribution will become more even.

Proposal II

- Refer to Source D. The installation of renewable energy systems involves high costs, so households that install these systems are generally with higher income. If tax incentives are provided to these households, it is likely that only the high-income households will benefit, resulting in an increase in their disposable income. While the low-income households will not benefit, income distribution will become more uneven.

Effects on aggregate output

Proposal I

- Increasing the electricity charges relief provided to eligible residential electricity accounts will cause their disposable income to increase. Private consumption expenditure will increase and aggregate demand will increase. In the short run, aggregate output will increase.
- Refer to Source D. Power companies are trying to make energy systems more efficient during the low-carbon transition, which may increase the productive capacity of Hong Kong. Long run aggregate supply will increase. In the long run, aggregate output will increase.

Proposal II

- Proposal II will encourage households to install renewable energy systems. Private consumption expenditure will increase and aggregate demand will increase. In the short run, aggregate output will increase.

Effects on government revenue and/or expenditure

Proposal I

- Increasing the electricity charges relief provided to eligible residential electricity accounts will increase government expenditure.
- Refer to Source D. Power companies investing in clean energy infrastructure during the low-carbon transition helps establish new industries, which may increase the government's profits tax revenue.

Proposal II

- Providing tax incentives will reduce government tax revenue.
- Proposal II will encourage households to install renewable energy systems, causing an increase in demand for renewable energy systems. The profits of renewable energy system producers may increase, which may increase the government's profits tax revenue.

Section C (16 marks)

12. (a) Verbal elaboration:

Inefficient. At Q_M , marginal benefit is higher than marginal cost and there is a deadweight loss.

(1 mark)

Illustrate in the diagram:

- correct position of marginal revenue

(1 mark)

- correct position of Q_M

(1 mark)

- correct position of P_M

(1 mark)

- correct position of deadweight loss

(1 mark)

(b) (i) Illustrate in the diagram:

- correct position of Q_S

(1 mark)

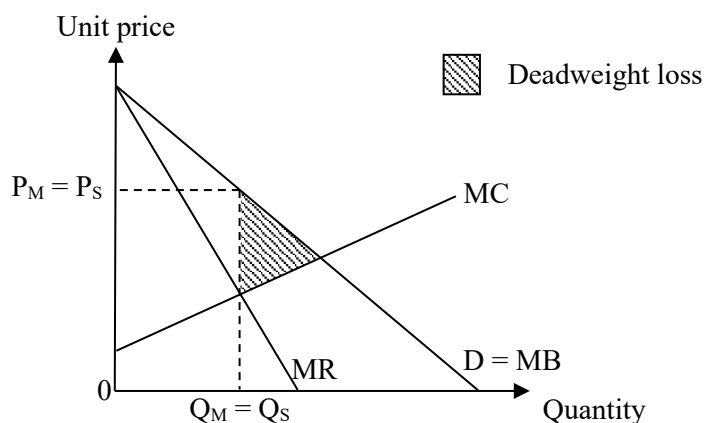
- correct position of P_S

(1 mark)

(ii) No, because neither demand nor marginal cost is affected by the provision of the subsidy.

Deadweight loss will remain unchanged.

(2 marks)



(c) No, because all consumers are entitled to the same price arrangement and there is no market segmentation.

(3 marks)

(d) (i) Refusal to supply essential facilities.

(1 mark)

A possible motive:

- Prevent competitors from entering the market to compete with it.

- Any reasonable answer

(Mark the **FIRST** point only)

1@, max: 1 mark

(ii) Possible losses:

- Consumers may have to pay a higher price.

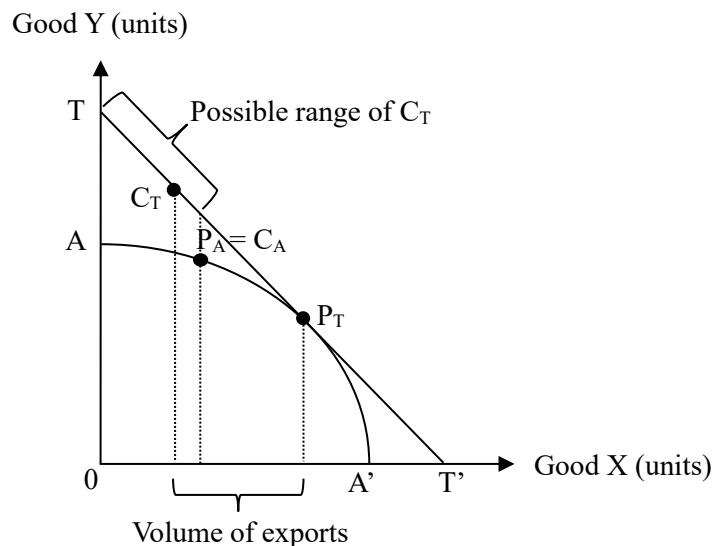
- Consumers will lack choices.

- Any reasonable answers

(Mark the **FIRST TWO** points only)

1@, max: 2 marks

13. (a) The marginal costs of producing both good X and good Y are increasing in Country A. (1 mark)
- (b) Verbal elaboration:
 Without trade, Country A can only consume its own production. (1 mark)
Illustrate in the diagram:
 - correct position of C_A = production point P_A (1 mark)
- (c) As the opportunity cost of producing good X in Country A (represented by the slope of the PPF at point P_A) is lower than the terms of trade of good X (represented by the slope of the international price line), Country A has a comparative advantage in producing good X. (3 marks)
- (d) (i) Verbal elaboration:
 - Country A can produce at a point which is unattainable from domestic production before trade. (1 mark)
Illustrate in the diagram:
 - correct position of production point after trade (P_T) (1 mark)
 - correct position of consumption point after trade (C_T) (1 mark)
- (ii) Illustrate in the diagram:
 - correct position of volume of exports (1 mark)



- (e) No, because the opportunity cost of producing good X and good Y will not be affected. (2 marks)
- (f) Possible reasons:
- Productivity may have increased due to the import of superior technology from trading partners, leading to higher per capita gross national income.
 - Import of foreign health products or technology may have increased life expectancy.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- 2@, max: 4 marks